# DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001-2015 Certified Institution)
Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru - 560 111. India

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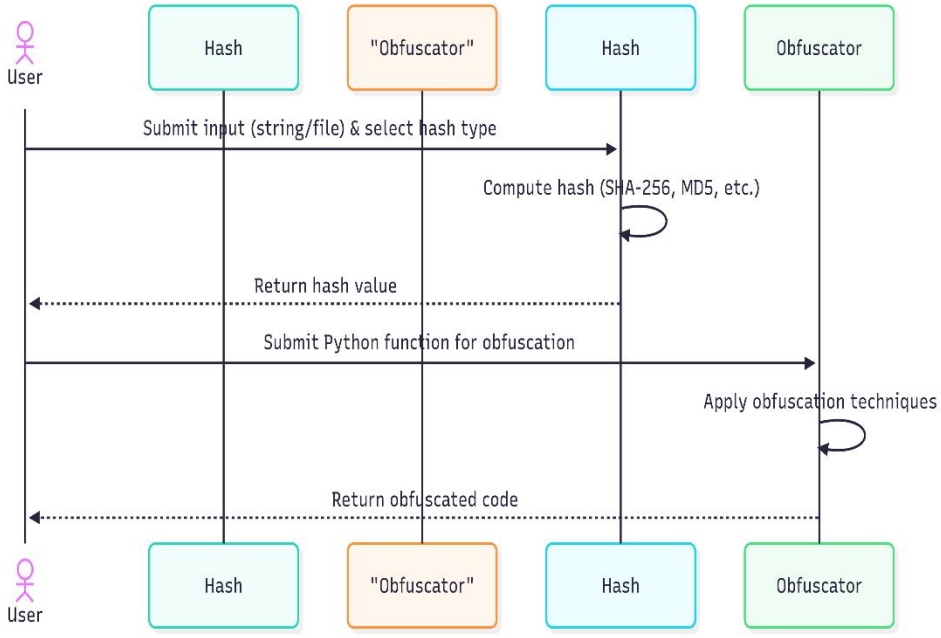
| DEPARTMENT | **ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING** | | | | |
| **Semester: VII** | **Course Title: Data Security & Privacy Lab** | | | **Course Code: 22AI73** | |
| PROGRAM QUESTION | Hash Functions and Obfuscation:a)Implement common hash functions (e.g., SHA-256, MD5) using Python's hashlib. Input: A string or file. Output: Its hash value. | | | | |
| STUDENT NAME | Udayaraj | | USN | 1DS22AI057 | |
| LAB MARKS | **Program Execution (10)** | **Source Code (10)** | **Result (5)** | **Innovation/ Extra Features (5)** | **Total** |
| | | | | | |
| TOOLS USED | Cursor | | | | |
| MAP COURSE OUTCOME | CO1 | | | | |
| MAP PROGRAM OUTCOME | PO2, PO3, PO5, PO9, PO11 | | | | |
| MAP PROGRAM SPECIFIC OUTCOME | PSO1, PSO2, PSO3 | | | | |
| UML / SEQUENCE DIGRAM |  | | | | |
| COURSE COORDINATOR | Dr ARUNA M G & Prof ENSTEIH SILVIA | | | | |

**Code for Program 7:**

7) Hash Functions and Obfuscation:a)Implement common hash functions (e.g., SHA-256, MD5) using Python's hashlib. Input: A string or file. Output: Its hash value.

b) Explore obfuscation techniques

```
import streamlit as st, hashlib, base64
def hash_text(text, algo): return hashlib.new(algo, text.encode()).hexdigest()
def hash_file(file, algo): return hashlib.new(algo, file.read()).hexdigest()
def obfuscate_code(code):
    encoded = base64.b64encode(code.encode()).decode()
    return f'import base64\nexec(base64.b64decode("{encoded}").decode())'
st.title("🔐 Hash & Code Obfuscator")
choice = st.sidebar.radio("Select Mode", ["Hash Generator", "Code Obfuscator"])
if choice == "Hash Generator":
    input_type = st.radio("Input type", ["Text", "File"])
    if input_type == "Text":
        txt = st.text_area("Enter text")
        if st.button("Generate Hash") and txt:
            st.write("MD5:", hash_text(txt, "md5"))
            st.write("SHA-256:", hash_text(txt, "sha256"))
    else:
        f = st.file_uploader("Upload file")
        if f and st.button("Generate File Hash"):
            st.write("MD5:", hash_file(f, "md5"))
            f.seek(0)
            st.write("SHA-256:", hash_file(f, "sha256"))
else:  # Code Obfuscator
    code = st.text_area("Paste Python code")
    if st.button("Obfuscate") and code.strip():
        st.subheader("🔎 Original Code"); st.code(code, language="python")
        st.subheader("🕵️ Obfuscated Code"); st.code(obfuscate_code(code), language="python")
```

# RESULT: